

Network Topologies

Network topology describes how computers and other devices are connected to each other in a network, either physically or logically.

1. Bus Topology

All devices are connected to a **single main cable** called the backbone.

- Simple and inexpensive
- Easy to install for small networks
- Failure of the main cable can affect the entire network
- Performance decreases as more devices are added

2. Star Topology

All devices are connected to a **central device**, such as a switch or hub.

- Easy to install and manage
- Failure of one device usually does not affect others
- Failure of the central switch can stop the network
- Common in modern LANs

3. Ring Topology

Each device is connected to **two neighboring devices**, forming a circular arrangement.

- Data travels around the ring
- Provides an organized data path
- Failure of one connection can disrupt communication in a basic ring
- Less common in modern networks

4. Mesh Topology

Devices are connected to **multiple or all other devices**.

- Highly reliable
- Provides multiple paths for data
- Expensive because it requires many connections
- Used where reliability is very important

5. Tree Topology

A combination of **star and bus topologies**, arranged in a hierarchical structure.

- Suitable for large networks
- Easy to expand
- Easier to manage by sections
- Failure of a higher-level connection can affect connected branches

6. Hybrid Topology

A combination of **two or more different topologies**, such as star + bus.

- Flexible and scalable
- Can be designed according to organizational needs
- More complex and expensive to maintain

7. Point-to-Point Topology

A **direct connection between two devices**.

- Simple and fast
- Easy to configure
- Common in direct communication links
- Connects only two endpoints